FIT3080 Artificial Intelligence
Practice Questions in MDPs and Reinforcement Learning

1. **[MDP and RL]** Consider a simple MDP with two states: S_1 and S_2, two actions: A and B, a discount factor γ of $1/2$, a reward function R given by:

$$R(s,a,s') = \begin{cases} 2 & \text{if } s = S_1, a = A, s' \in \{S_1, S_2\} \\ 3 & \text{if } s = S_1, a = B, s' \in \{S_1, S_2\} \\ -2 & \text{if } s = S_2, a = A, s' \in \{S_1, S_2\} \\ -3 & \text{if } s = S_2, a = B, s' \in \{S_1, S_2\} \end{cases}$$

The transition function is specified by the following table:

s	a	s'	$\Pr(s' \mid s, a)$
S_1	A	S_1	$4/5$
S_1	A	S_2	$1/5$
S_1	B	S_1	$1/2$
S_1	B	S_2	$1/2$
S_2	A	S_1	$1/2$
S_2	A	S_2	$1/2$
S_2	B	S_1	$2/3$
S_2	B	S_2	$1/3$

The Bellman equations appear below:

$$V^*(s) = \max_a Q^*(s,a)$$
$$Q^*(s,a) = \sum_{s'} T(s,a,s') \Big[R(s,a,s') + \gamma V^*(s') \Big]$$
$$V^*(s) = \max_a \sum_{s'} T(s,a,s') \Big[R(s,a,s') + \gamma V^*(s') \Big]$$

(a) Suppose that after i iterations of Value Iteration, the values are now as shown in the following table. Perform a single round of value iteration, filling in the resulting state values, on the basis of the provided initial values V_i (rather than starting from zero state values).

s	V_i	V_{i+1}
S_1	1	
S_2	-1	

(b) Perform a single round of Q-value iteration, filling-in the resulting Q-values in the following table. Use the specified initial value function Q_i, rather than starting from all zero Q-values. Only compute the entries not labeled "skip".

s	a	Q_i	Q_{i+1}
S_1	A	2	
S_1	B	1	
S_2	A	-3	skip
S_2	B	-2	skip

(c) Suppose that we did not know the transition function T. We still need to find an optimal policy, so we make use of Q-learning, with a learning rate $\alpha = 1/4$ and a discount factor of $\gamma = 1/2$.
We limit our learning episodes to at most 5 steps, and our first episode is:

$$[S_1, A, S_1, A, S_2, B, S_2, A, S_1, B, S_1]$$

Initializing the $Q(s, a)$ to zero, and using the rewards from item (a), what would be the new values for $Q(s, a)$ after observing this episode?
The Q-learning equation is:

$$Q(s_t, a_t) \leftarrow (1 - \alpha) \cdot Q(s_t, a_t) + \alpha \cdot \left(r_t + \gamma \cdot \max_a Q(s_{t+1}, a)\right)$$

2. **[MDP]** To decide whether to save or invest in stocks, we are given the following information. If you have \$1000+,

- 1 year of saving has a 50% chance of gaining \$1000, and a 50% chance of maintaining your current amount.
- 1 year of investing has a 75% chance of gaining \$2000, and a 25% chance of losing \$1000.

You currently have \$1000, and a goal to save at least \$4000. Achieving your goal has a reward of 8, while losing all your money (\$0) has a reward of -16. In both cases you have to stop investing. In addition, reaching every other state has a reward of -1. You do not use discounting ($\gamma = 1$).

(a) Express the problem stated above as a Markov Decision Process. What are the States, Actions, Transitions and Rewards?

(b) What is the start state? What are the terminal states?

(c) The Bellman equations are as follows:

$$V^*(s) = \max_a Q^*(s, a)$$
$$Q^*(s, a) = \sum_{s'} T(s, a, s') \Big[R(s, a, s') + \gamma V^*(s')\Big]$$
$$V^*(s) = \max_a \sum_{s'} T(s, a, s') \Big[R(s, a, s') + \gamma V^*(s')\Big]$$

Write down the Bellman equation for the optimal value $V^*(s)$ of the starting state in terms of the values $V^*(s')$ of the states reachable from the initial state, simplifying as much as possible.

(d) The formula for the policy improvement step is

$$\pi_{k+1}(s) = \arg\max_a \sum_{s'} T(s,a,s')[R(s,a,s') + \gamma V^{\pi_k}(s')]$$

Given policy π with the following values $V^\pi(\$1000) = -2$, $V^\pi(\$2000) = 0$ and $V^\pi(\$3000) = 4.5$, what would be the policy $\pi'(s)$ for the start state \$1000 after a policy improvement step?

3. **[MDP and RL]** Consider an autonomous robot which can either move fast or slow in any time step. Moving fast generally gives a reward of +2, while moving slow gives a reward of only +1. However, the robot must also take into account its internal temperature, which can be either hot or ok . Driving slow tends to lower the temperature, while driving fast tends to raise it. If the robot is hot, there is a danger if it overheating, at which point it must stop, cool down, and make repairs. The MDP transitions and rewards are specified as follows:

s	a	s'	$T(s,a,s')$	$R(s,a,s')$
ok	slow	ok	1	+1
ok	fast	ok	0.5	+2
ok	fast	hot	0.5	+2
hot	slow	ok	1	+1
hot	fast	hot	0.5	+2
hot	fast	ok	0.5	-10

The Bellman equations appear below:

$$V^*(s) = \max_a Q^*(s,a)$$
$$Q^*(s,a) = \sum_{s'} T(s,a,s')\Big[R(s,a,s') + \gamma V^*(s')\Big]$$
$$V^*(s) = \max_a \sum_{s'} T(s,a,s')\Big[R(s,a,s') + \gamma V^*(s')\Big]$$

(a) Run two rounds of value iteration in the table below, using a discount of 0.8. You may skip the entry labeled "skip".

s	V_0	V_1	V_2
ok	0		
hot	0		skip

(b) Run Q-learning with a discount of .8 and a learning rate of 0.5, using the transition samples below. Do not copy over q-values which have not changed in a given step. Assume the agent experiences the samples:
ok, fast, hot, reward +2, calculate Q_1
hot, fast, ok, reward -10, calculate Q_2
ok, slow,ok, reward +1, calculate Q_3

s	a	Q_0	Q_1	Q_2	Q_3
ok	**slow**	0			
ok	**fast**	0			
hot	**slow**	0			
hot	**fast**	0			

The Q-learning equation is:

$$Q(s_t, a_t) \leftarrow (1-\alpha) \cdot Q(s_t, a_t) + \alpha \cdot \left(r_t + \gamma \cdot \max_a Q(s_{t+1}, a)\right)$$

4. [**MDP**] Suppose you are given a simple 3-state MDP, specified with the following transition function T and the reward function R. In the table provided below, run the value iteration algorithm until the policy converges. Use the discount factor $\gamma = 0.5$. When choosing the maximum valued action for each state, break ties by choosing left before right.

Reward Function

s	$R(s)$
a	2
b	4
c	-10

Transition Model

s_{t-1}	a_{t-1}	$P(s_t = a \mid s_{t-1}, a_{t-1})$	$P(s_t = b \mid s_{t-1}, a_{t-1})$	$P(s_t = c \mid s_{t-1}, a_{t-1})$
a	left	0	0.2	0.8
a	right	0	0.8	0.2
b	left	0.8	0	0.2
b	right	0.2	0	0.8
c	left	0.2	0.8	0
c	right	0.8	0.2	0

Value Iteration

Iteration	$V(a)$	$V(b)$	$V(c)$	$\pi(a)$	$\pi(b)$	$\pi(c)$
1	0	0	0	left	left	left
2						
3						
4						
5						
6						
7						